

① What is the purpose of this Algorithm in program?

Answer: This program uses hill climbing algorithm to find the maximum value of a mathematical function

Function:

$$f(x) = -x^2 + 5$$

② Why it is called 'Hill Climbing'?

Because the algorithm behaves like a person climbing a hill
Check nearby points Move toward a higher point
Stops when no nearby point is higher

③ What does this function do?

```
def objective(x):
```

```
    return -x[0]**2 + 5
```

This function algorithm want to optimize it receive an input x and returns a Score/Value

Example:

$$x=0 \rightarrow \text{Value} = 5x - 2 \rightarrow \text{Value } 1$$

④ Why do we use $x[0]$ instead of x ?

x is stored as Numpy array
 $\text{np.array}([2.0])$ So $x[0]$

finds value inside array 5.

⑤ What does this function Shapes?

$$y = -x^2 + 5$$

forms an upside-down parabola
Highest point $x=0$ lower values as
we move away from 0.

⑥ Why is the max at $x=0$?

Because $f(0) = 5$
give Highest Value among others

⑦ What does generate-neighbours() do?

it creates nearby possible solution

⑧ Why are neighbours important?

Hill climb looks at nearby neighbors

⑨ What is Step-Size?

This control how far the algorithm
moves each step

⑩ What does `current = np.array([initial])` do?

Storing starting point as a Numpy array

(11) Why do we need a starting point?
it cannot be start nowhere it need initial state.

(12) What does `current-eval-obj (current)` do?
it evaluate how good the current Sol is

(13) Why do we use a loop?

Hill Climbing improve Solution repeatedly.

(14) What happens inside `neighbour-evals`?

The algorithm calculates Scores for neighbors.

(15) What does `np.argmax()` do?

it finds the index of the highest Value.

(16) Why this Condition? if `neighbour-evals (best-idx) > current-eval`.

Hill climbing only move if the new Solution is better.

(17) What happens when algorithm moves?

(18) Why does algorithm stops?

(19) What is Converged here?

(20) What is complete flow?